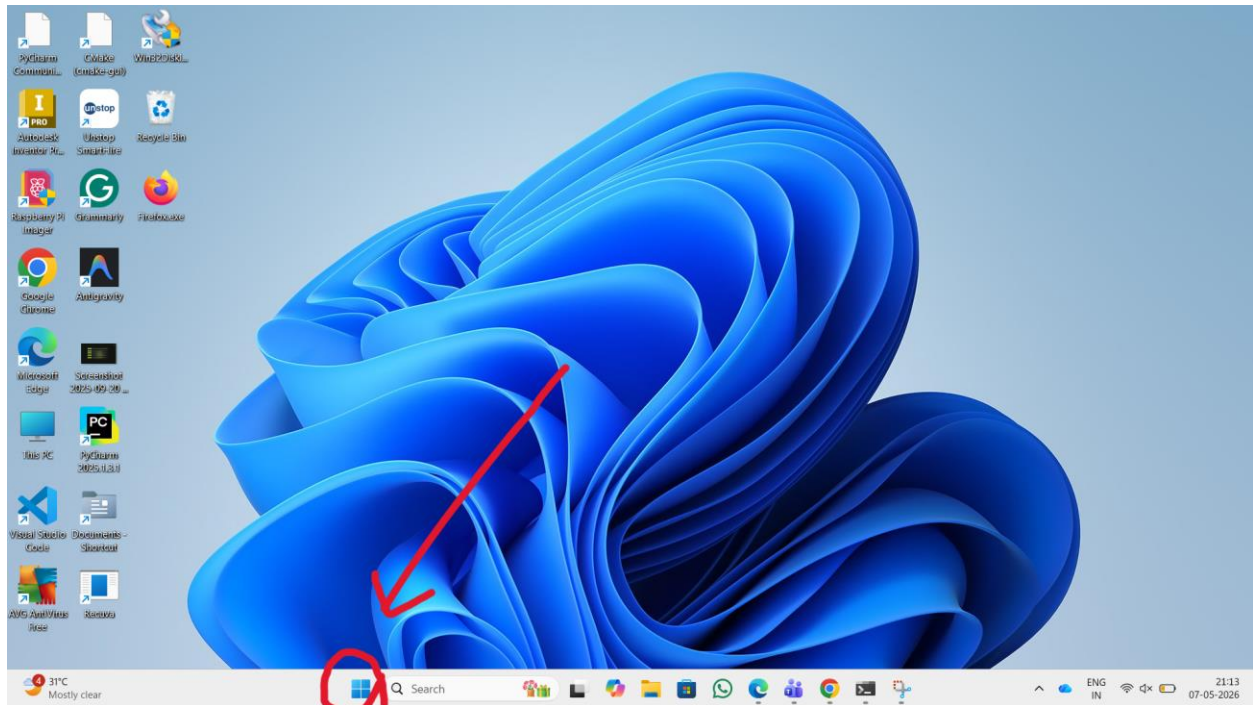
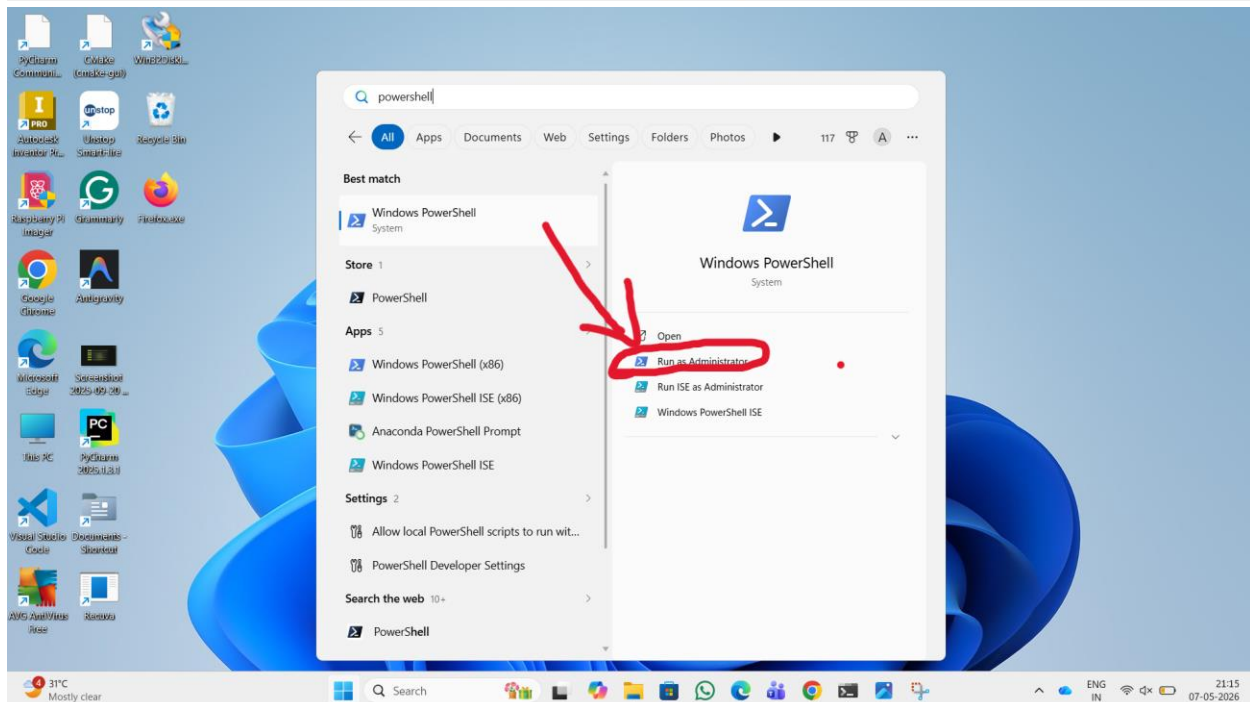
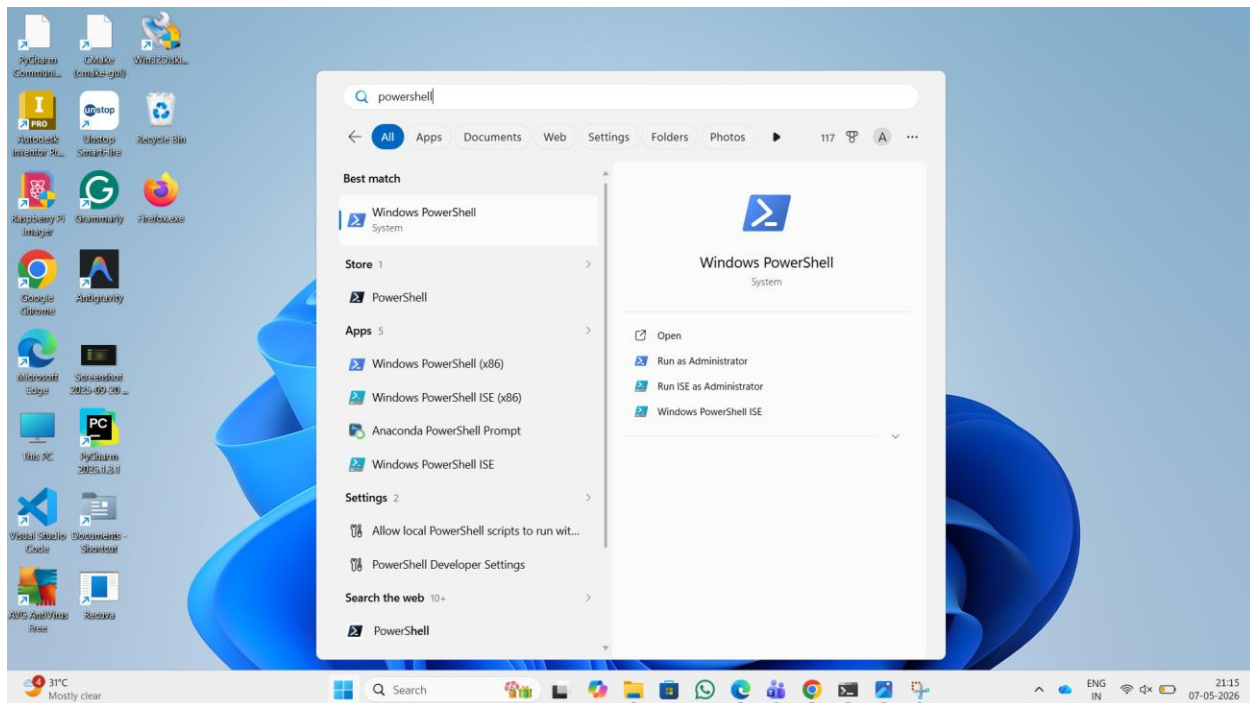


Ubuntu and Hadoop Installation on Windows Device

-By Aditya Nikam(BE AI & DS)

1)Step 1:-On your Windows Laptop/Computer click on Start Menu and Type “Powershell” and on right hand side of start menu you will see an option of “Run as administrator” after you typed “Powershell” so click on “Run as administrator”

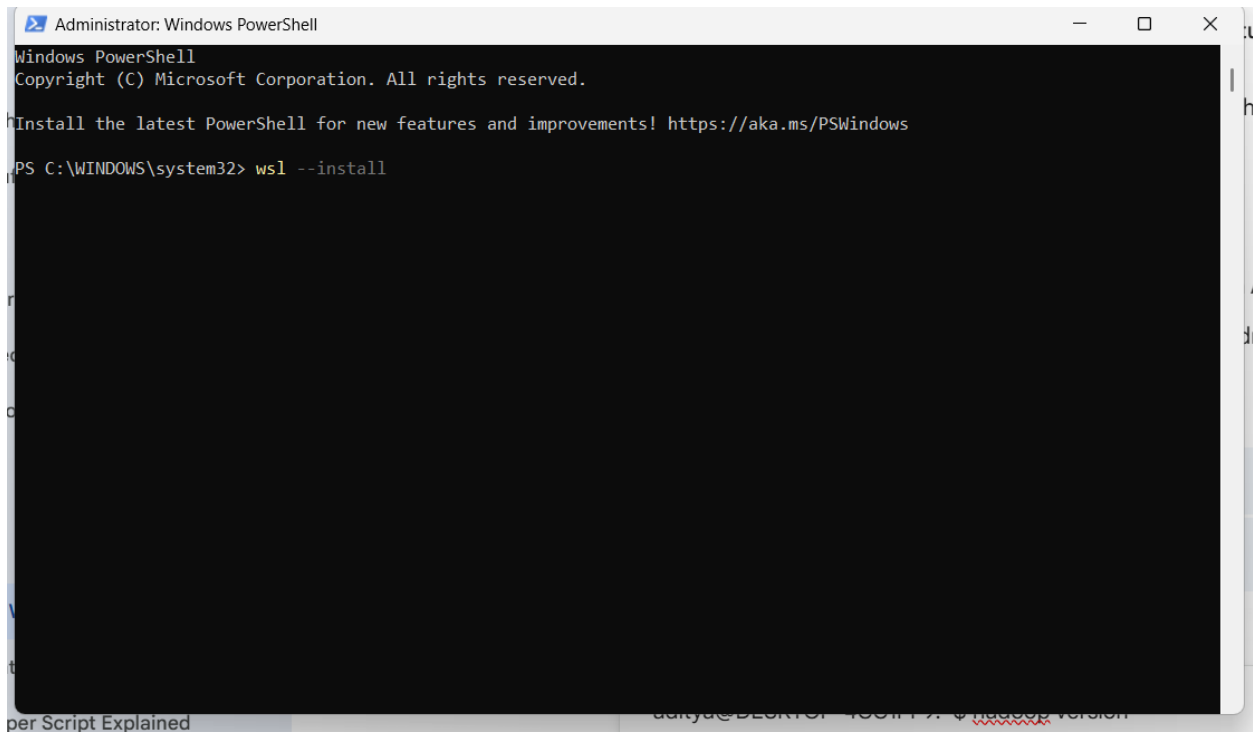




Step 2:-Now it will open a pop up after clicking Run as administrator. It that pop up it will ask you do you want to this app to make changes in your device.Click on Yes.

Step 3:-Now type the following command in the Powershell:

```
wsl --install
```



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> wsl --install
```

Now if the output of the command is like below:

```
Downloading: Windows Subsystem for Linux 2.6.3
Installing: Windows Subsystem for Linux 2.6.3
Catastrophic failure
PS C:\WINDOWS\system32>
```

Then type the below command inside the Powershell itself

```
dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart
```

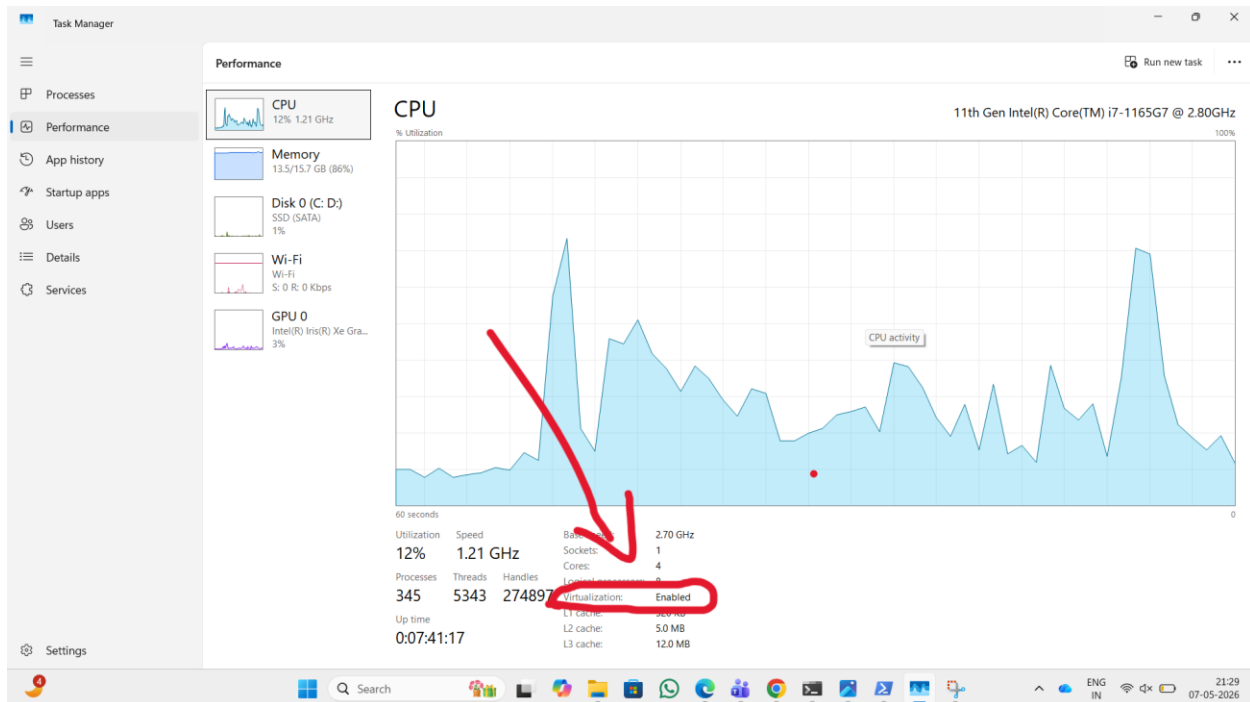
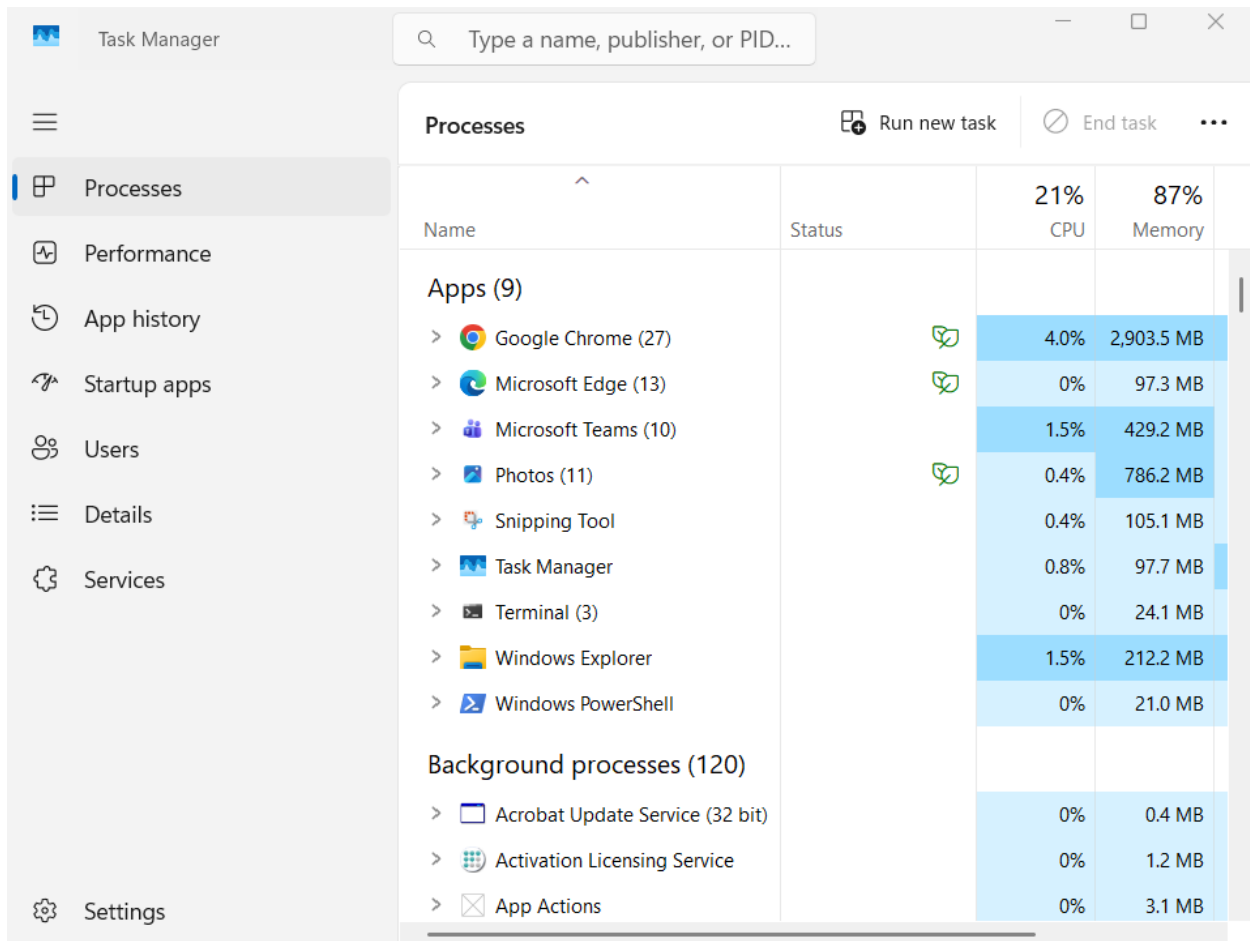
Now you will get the output as “The operation completed successfully”.Now type one more command like below:-

```
dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart
```

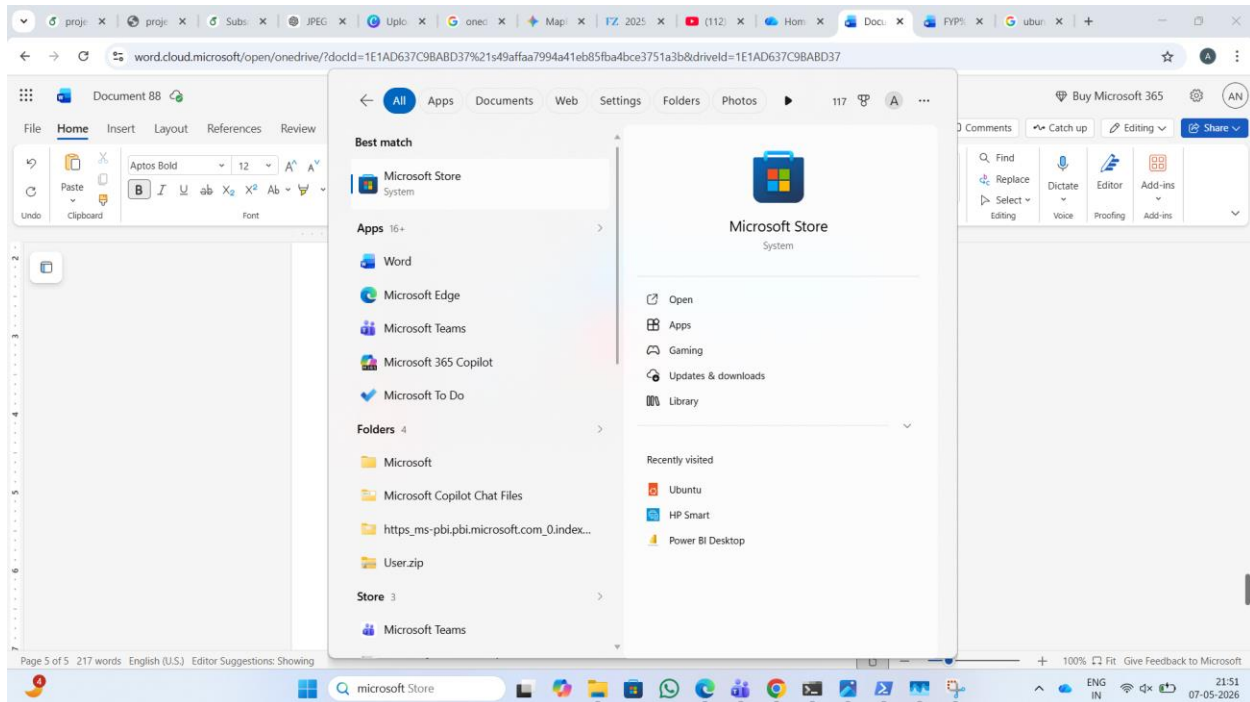
Now again you will get the output as “The operation completed successfully”

Step 4:-Restart your laptop/computer now

Step 5:-Now press Ctrl + Shift + Esc to open Task Manager and after opening Task Manager Click on “Performance” tab.There you will see Virtualization as enabled



Step 6:-Now click on Start Menu and Type “Microsoft Store” and open it



Step 7:-Now inside Microsoft Store type “Ubuntu” in search bar and then in results which you will get after pressing Enter click on Ubuntu to Install it .After Installing open Ubuntu you will see that one black color terminal window will pop up and content which is given below will be printed:

Windows Subsystem for Linux must be updated to the latest version to proceed. You can update by running 'wsl.exe --update'.

For more information please visit <https://aka.ms/wslinstall>

Press any key to install Windows Subsystem for Linux.

Press CTRL-C or close this window to cancel.

This prompt will time out in 60 seconds.

Now press any key on your keyboard to update Windows Subsystem for Linux and then let the Installation/Updating finish

Now below content will be printed:

Installing, this may take a few minutes...

Please create a default UNIX user account. The username does not need to match your Windows username.

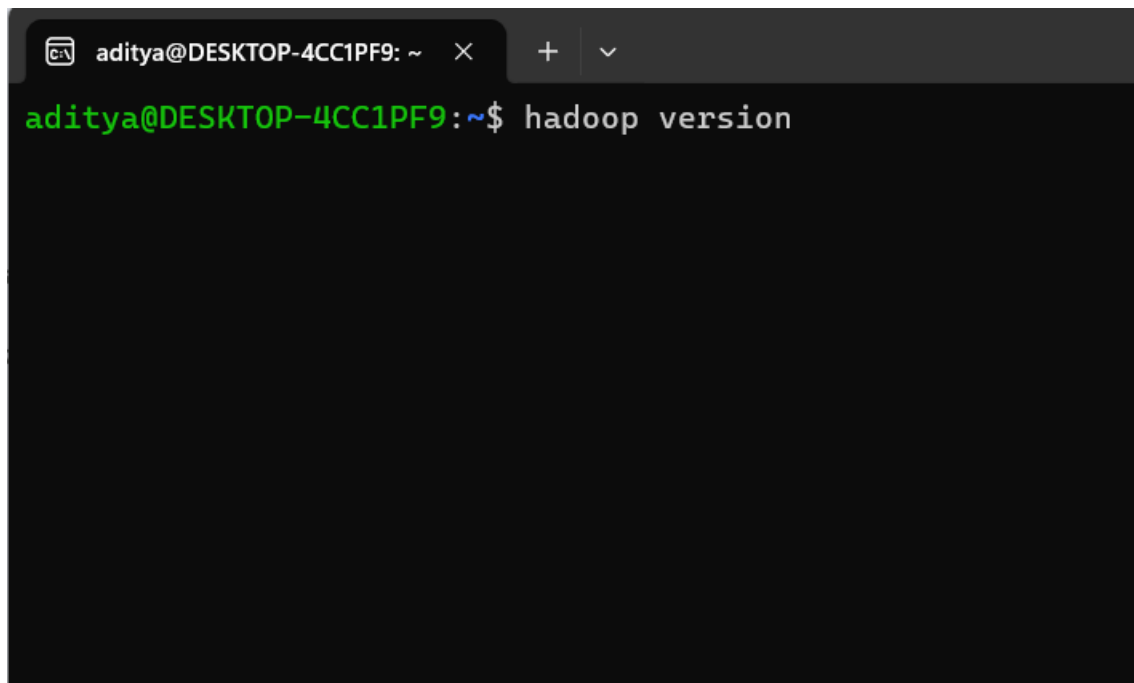
For more information visit: <https://aka.ms/wslusers>

Enter new UNIX username:

So enter your UNIX Username it can be anything like “aditya’,”student” etc.Then on next line it will ask you enter password.So type the password and note that while typing the password you will not see the characters on that screen for security reasons but the characters will be typed.After typing password it will ask to re enter the password so re enter the password

Step 8:-Now type below command in the Ubuntu Terminal after re-entering the password:

hadoop version

A screenshot of a terminal window. The title bar shows 'aditya@DESKTOP-4CC1PF9: ~'. The terminal prompt is 'aditya@DESKTOP-4CC1PF9:~\$' and the command 'hadoop version' has been entered. The terminal background is black with green text for the prompt and white text for the command.

- **If it is installed:** You will see a block of text telling you the version number (e.g., Hadoop 3.3.6).
- **If it is NOT installed:** You will see an error message like `hadoop: command not found`

If hadoop is not installed then follow the below steps:-

Step 9:-Type the below command on the Ubuntu Terminal:-

sudo apt update

```
aditya@DESKTOP-4CC1PF9:~$ sudo apt update
```

Step 10:-Type the below command on the next line in Ubuntu Terminal

```
sudo apt install openjdk-8-jdk -y
```

```
aditya@DESKTOP-4CC1PF9:~$ sudo apt install openjdk-8-jdk -y
```

(It will ask for your Ubuntu password you created in Step No.7. Remember it is invisible as you type it.)

Step 11:-Type the below command on new line in Ubuntu Terminal

```
wget https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz
```

This command will download Hadoop.Let the downloading finish it may take some minutes.You will see output like below:-

```
aditya@DESKTOP-4CC1PF9:~$ wget
```

```
https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz
```

```
--2026-05-07 15:32:06-- https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz
```

```
Resolving downloads.apache.org (downloads.apache.org)... 88.99.208.237, 95.216.224.44, 2a01:4f8:10a:39da::2, ...
```

```
Connecting to downloads.apache.org (downloads.apache.org)|88.99.208.237|:443... connected.
```

```
HTTP request sent, awaiting response... 200 OK
```

```
Length: 730107476 (696M) [application/x-gzip]
```

```
Saving to: 'hadoop-3.3.6.tar.gz'
```

```
hadoop-3.3.6.tar.gz
```

```
100%[=====>] 696.28M 2.66MB/s  
in 8m 38s
```

2026-05-07 15:41:22 (1.34 MB/s) - 'hadoop-3.3.6.tar.gz' saved [730107476/730107476]

This command extracts the files

Step 12:-Now type the below command:

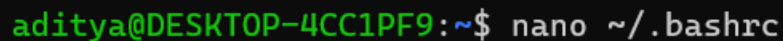
```
sudo mv hadoop-3.3.6 /usr/local/hadoop
```

We moved the extracted files to a proper system folder (/usr/local/hadoop).

Step 13:-Restart your laptop/computer and then after restarting open Ubuntu again.

Step 14:-After opening Ubuntu type below command in Ubuntu Terminal:-

```
nano ~/.bashrc
```



```
aditya@DESKTOP-4CC1PF9:~$ nano ~/.bashrc
```

Step 15:- Use your Down Arrow Key on your keyboard to scroll to the very bottom of the file. Paste this exact text at the bottom:

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64 export
HADOOP_HOME=/usr/local/hadoop export
PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
```

Step 16:-Press Ctrl + O then Enter and then Ctrl + X.Now type the below command:

```
source ~/.bashrc
```

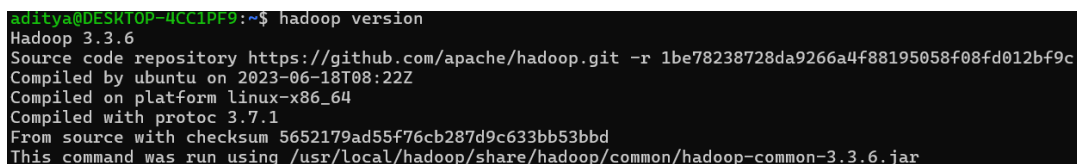


```
aditya@DESKTOP-4CC1PF9:~$ source ~/.bashrc
```

Step 17:-Now type the below command on new line:

```
hadoop version
```

Step 18:- You will see the output like below.This shows that Hadoop is now installed



```
aditya@DESKTOP-4CC1PF9:~$ hadoop version
Hadoop 3.3.6
Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f08fd012bf9c
Compiled by ubuntu on 2023-06-18T08:22Z
Compiled on platform linux-x86_64
Compiled with protoc 3.7.1
From source with checksum 5652179ad55f76cb287d9c633bb53bbd
This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar
```